

Deconfounding Multimodal Affective Computing via Minimal Global Causal Representations

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Abstract

Multimodal Affective Computing (MAC) mimics the human brain’s ability to integrate diverse modalities, significantly enhancing human-computer interaction. While multimodal representation learning is effective when training and testing data share the same distribution, its performance often drops sharply when encountering out-of-distribution (OOD) data. This limitation makes many existing MAC methods unsuitable for real-world applications. To address this, we propose the Causal Multimodal Information Bottleneck (CaMIB) framework to learn generalized multimodal representations under distribution shifts. Specifically, we first construct an informative latent space by applying the information bottleneck principle at the unimodal level, purifying each input through variational approximation to capture task-relevant features. We then employ a parameterized mask generator to split the fused multimodal representation into causal and confounding branches. To disentangle these components, we apply causal intervention to stratify the confounding representations, pairing causal features with diverse confounding counterparts to stabilize the mapping between causal estimation and prediction. Finally, an instrumental variable constraint is introduced to ensure the global consistency of the learned causal representations. Extensive experiments on multimodal sentiment analysis, humor detection, and sarcasm detection, along with OOD test sets, demonstrate the effectiveness of CaMIB. Theoretical and empirical analyses further highlight its interpretability.

CCS Concepts

• Information systems → Multimedia information systems.

Keywords

Multimodal Affective Computing, Global Causal Representation, Information Bottleneck, Out-of-distribution Generalization

1 Introduction

Multimodal Affective Computing (MAC) is a rapidly advancing field that aims to mimic the human ability to perceive emotions through multiple sensory channels. By integrating information from spoken language, facial expressions, and acoustic cues, MAC enables a more comprehensive understanding of human emotional states and sentiment intensity. Given its ability to bridge the gap between human perception and machine intelligence, MAC enables a wide range of applications, including dialogue systems [75], intelligent tutoring systems [39], and social media analysis [73].

Multimodal representation learning aims to extract informative signals from multiple modalities while capturing both their shared characteristics and heterogeneity [24]. It has achieved notable success across diverse MAC tasks such as Multimodal Sentiment Analysis (MSA) [71], Multimodal Humor Detection (MHD) [14],

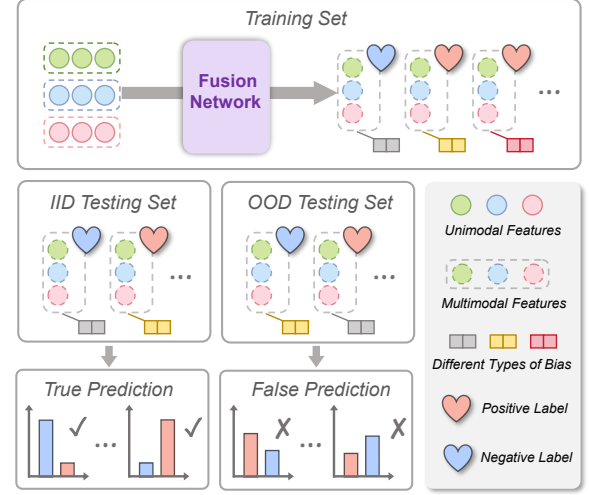


Figure 1: An illustration of generalization failure under distribution shifts.

and Multimodal Sarcasm Detection (MSD) [5]. Early studies primarily relied on tensor fusion [69, 70] to model cross-modal interactions. To enable more fine-grained integration, subsequent approaches shifted toward cross-modal attention [6, 52, 53] and Transformer-based architectures [19, 42, 51]. More recently, contrastive learning [25, 30, 34, 67] and self- and weakly supervised paradigms [11, 28, 68] have been introduced to bridge the modality gap, facilitating more robust feature alignment.

Despite the enormous success, these methods heavily rely on the I.I.D. assumption, which assumes that training and test data are drawn from the same distribution. However, in real-world scenarios, distribution shifts are prevalent and often unavoidable due to uncontrolled data generation processes [4]. A key limitation is that existing approaches frequently overlook causal patterns, instead relying on spurious correlations within multimodal data. Since deep neural networks tend to exploit such shortcuts for prediction, most current models fail to generalize to out-of-distribution (OOD) data [46]. Figure 1 illustrates such distribution shifts and their adverse impact on generalization performance. Consequently, learning robust multimodal representations and developing methods capable of OOD generalization is essential, especially for high-stakes applications such as medical diagnosis [62].

Encouragingly, causal inference [37] offers a promising direction for addressing this challenge, as it enables the identification of underlying causal relationships even in biased observational data. However, applying causal inference to MAC tasks remains non-trivial and presents two key challenges. ❶ *How can causal and confounding substructures be reliably identified in biased datasets?*

When the test distribution deviates from the training distribution, models tend to exploit spurious correlations, leading to unreliable predictions [44, 66]. Existing causal approaches often address this issue by predefining specific bias types and mitigating them via counterfactual reasoning [17, 18, 46, 65] or causal interventions [45, 61, 63]. However, such methods are typically limited to local or narrowly defined biases and lack the ability to distinguish causal from confounding substructures at a global level. **Q How can causal substructures be extracted from entangled multimodal inputs?** From a statistical perspective, causal substructures are governed by the global properties of multimodal data rather than individual modalities or local features alone [9]. Effectively capturing them thus requires modeling both complex inter-modal interactions and intra-modal dependencies.

To tackle these challenges, we first formulate a Structural Causal Model (SCM) tailored to MAC, where spurious correlations induced by redundant information are explicitly modeled as confounders, moving beyond pre-defined bias assumptions. Since these confounders can mislead inference and result in biased predictions, we propose the Causal Multimodal Information Bottleneck (CaMIB) framework to mitigate their effects. Inspired by recent advances in information bottleneck (IB) methods [29, 31, 59], our approach bridges IB with causal theory. IB is employed to learn compact representations by filtering out irrelevant information [49]. However, IB-based objectives typically aim to maximize the mutual information between inputs and targets, which may be more prone to capturing spurious correlations. To address this issue, we further incorporate causal deconfounding to eliminate the influence of such factors, leading to more robust representations.

Specifically, we construct an informative latent space by applying the IB principle at the unimodal level, learning compact representations via a variational approximation to retain task-relevant features while suppressing redundant noise. We then design a parameterized mask generator to partition the fused multimodal representation into causal and confounding branches, with parameters shared across feature dimensions. To disentangle these components, we perform causal interventions to stratify with respect to confounders. By pairing causal representations with diverse confounders, we generate stratified samples with reduced correlations, thereby enforcing an invariant relationship between causal patterns and predictions under variations in confounders and distribution shifts. Finally, we introduce an instrumental variable mechanism based on self-attention to capture inter-modal and token-level dependencies, while ensuring global causal consistency.

Our main contributions are summarized as follows:

- We formalize a SCM tailored for MAC that treats redundant information and spurious correlations as confounders, offering a novel perspective on multimodal bias.
- We propose CaMIB, a deconfounding framework that bridges the IB principle and causal theory to extract robust global multimodal causal representations while filtering out task-irrelevant noise.
- Extensive experiments on multiple MAC tasks and OOD test sets show that CaMIB consistently outperforms existing methods. Theoretical analysis further supports the interpretability of CaMIB.

2 Related Work

2.1 Information Bottleneck

The Information Bottleneck (IB) formulates representation learning within an information-theoretic framework, where task-relevant representations are characterized as a trade-off between feature compression and predictive power. It was first proposed in [49] and later extended to deep learning by [50]. Subsequently, the Variational Information Bottleneck (VIB) [2] bridges IB and deep learning by enabling efficient approximation via stochastic variational inference. Owing to its strong interpretability in representation learning and generalization analysis, IB has attracted significant attention across various domains, including computer vision [48], reinforcement learning [12], and natural language processing [56]. In this work, we adopt the IB principle to construct an informative latent space at the unimodal level by purifying each input through variational approximation, thereby capturing task-relevant features while filtering out irrelevant information.

2.2 Causal Inference

Moving beyond surface-level correlations, causal inference [37] provides a principled statistical framework for uncovering the underlying causal relationships among observed variables. Driven by recent advances in machine-learning-based causal demystification [7, 36, 57, 76], existing research in this area can be broadly categorized into two main directions: causal intervention [26, 61] and counterfactual reasoning [46, 65]. Causal intervention seeks to break the natural dependencies between variables by actively manipulating independent variables, thereby mitigating the influence of confounding factors [38]. In contrast, counterfactual reasoning focuses on hypothetical scenarios by examining how observed variables would behave under alternative treatment conditions [38]. In this work, we tackle confounding effects through causal intervention, as it offers greater flexibility for complex multimodal settings.

2.3 Multimodal Affective Computing

Given the inherent heterogeneity of multimodal data, a central objective in MAC research is to develop robust fusion mechanisms for learning high-quality joint representations [40, 78]. For example, tensor fusion [69], as a representative of early studies, focuses on modeling complex inter-modal interactions. Beyond architectural design, recent approaches incorporate information-theoretic and statistical constraints—including divergence-based objectives [43], correlation modeling [47], and information bottleneck principles [31]—to regulate cross-modal information flow and suppress task-irrelevant noise [59]. More recently, the field has witnessed the rise of large-scale pre-training paradigms [21]. Transformer-based backbones, equipped with multimodal pre-training objectives [19, 42, 51] or further enhanced through ensemble strategies [27], have demonstrated strong performance in modeling both intra- and inter-modal interactions. In parallel, self-supervised and contrastive learning frameworks have also emerged as popular approaches for leveraging unlabeled data to learn more discriminative and transferable multimodal representations [11, 28, 34, 67].

Among deep learning approaches for MAC, the methods most closely related to ours can be grouped into two categories. **First**, a

series of multimodal information bottleneck (MIB)-based methods aim to suppress redundancy and noise in both unimodal and multimodal representations. For instance, MIB [31] introduces three variants—early-fusion (E-MIB), late-fusion (L-MIB), and complete framework (C-MIB). ITHP [59] designates a primary modality and leverages the remaining modalities as auxiliary detectors to extract salient information. KAN-MCP [29] further integrates interpretable cross-modal modeling with DRD-MIB-based feature compression and denoising, thereby producing more discriminative representations while alleviating modality imbalance. **Second**, several methods grounded in causal theory are closely related. CLUE [46] proposes a counterfactual framework to capture the direct effect of the textual modality via an auxiliary text model. GEAR [45] introduces a generalized mean absolute error loss to disentangle robust and biased features. MulDeF [17] addresses label bias through causal intervention while mitigating multimodal bias via counterfactual reasoning.

Although these methods achieve strong performance, MIB-based approaches typically maximize mutual information between multimodal inputs and labels without discrimination, potentially capturing spurious correlations and overlooking dataset biases. Meanwhile, most causal approaches focus on single modalities or specific modality pairs, or rely on explicitly labeled bias types that require extensive domain expertise [35], limiting their applicability to complex multimodal data with implicit biases. In contrast, we propose a general and flexible deconfounding approach that performs causal interventions directly on fused multimodal representations without predefined bias types. Moreover, our method incorporates global causal structure during disentanglement, enhancing model generalization in complex multimodal scenarios.

3 Methodology

In this section, we begin by analyzing how multimodal representations are constructed for prediction in MAC from a causal perspective. Based on this causal view, we then present CaMIB, a novel framework that leverages causal intervention to learn generalized multimodal representations robust to distribution shifts. Finally, we provide a theoretical analysis of the proposed approach.

3.1 Preliminaries

MAC aims to infer a speaker’s emotional state—including sentiment scores, affective categories, and behavioral intentions—from a video segment comprising acoustic (a), visual (v), and textual (t) modalities. Formally, the input feature sequences are denoted as $U_m \in \mathbb{R}^{T_m \times d_m}$ for $m \in \{a, v, t\}$, where T_m and d_m represent the sequence length and feature dimensionality, respectively. Since different modalities may have inconsistent temporal resolutions, we follow prior work [29, 31, 59] and employ a 1D temporal convolutional layer to align them into a shared sequence length. The aligned features are denoted as $\tilde{U}_m \in \mathbb{R}^{l \times d_m}$, where l is the unified sequence length. For each modality, a dedicated encoder further maps the aligned sequences into a shared latent space, yielding unimodal representations $X_m \in \mathbb{R}^{l \times d}$, where d is the feature dimensionality. The objective is to learn a mapping from the joint set $\{X_a, X_v, X_t\}$ to the target affective label Y .

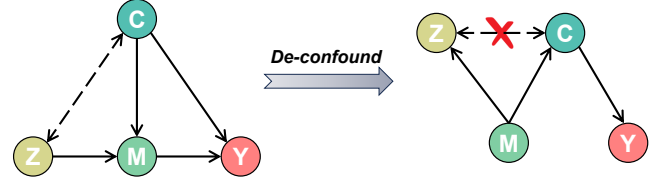


Figure 2: Comparison of SCMs. The left panel shows the causal graph of traditional MAC models, while the right panel illustrates the CaMIB framework, which utilizes the *do*-operator [37] to deconfound the multimodal fusion process.

3.2 A Causal View on MAC

To analyze causal influences in MAC tasks, we construct a causal graph to formalize the relationships among key variables. We adopt the standard notation of the structural causal model (SCM) framework [37]. Specifically, a causal graph $\mathcal{G} = \{\mathcal{N}, \mathcal{E}\}$ is a directed acyclic graph, where nodes $\mathcal{N}$ denote variables and edges $\mathcal{E}$ represent causal effects. We consider four variables (Figure 2): unobserved causal variables C , shortcut variables Z , multimodal representations M , and labels Y . This formulation is general and independent of specific model implementations. The causal dependencies are defined as follows:

- **Link:** $C \rightarrow M \leftarrow Z$. The multimodal representation M is generated from latent variables C and Z . Here, C captures task-relevant semantics (e.g., context and emotion), while Z encodes shortcut signals such as high-frequency words correlated with labels.
- **Link:** $C \rightarrow Y$. C is the direct causal parent of the label Y .
- **Link:** $M \rightarrow Y$. Since M depends on both C and Z , models may capture not only the true causal effect $C \rightarrow Y$ but also the spurious path $Z \rightarrow M \rightarrow Y$, leading to biased predictions.
- **Link:** $C \leftarrow - \rightarrow Z$. C and Z are correlated via unobserved confounders, making it difficult to disentangle causal and spurious factors.

According to d -connection theory [37], two variables are dependent if there exists at least one unblocked path. As shown in Figure 2, Z and Y are connected through two such paths: ① $Z \rightarrow M \rightarrow Y$ and ② $Z \leftarrow - \rightarrow C \rightarrow Y$, both inducing spurious correlations. To address this, we block both paths. For path ①, we disentangle C and Z in M and use only C for prediction. For path ②, since $C \rightarrow Y$ cannot be altered, we enforce independence between C and Z , thereby blocking the $C \leftarrow - \rightarrow Z$ connection.

3.3 Model Overview

Motivated by the above causal analysis, we propose CaMIB for deconfounding. As illustrated in Figure 3, our model consists of four main steps. 1) The IB module removes unimodal noise irrelevant to prediction, producing compact intermediate representations. 2) These representations are stacked across modalities, and a self-attention module captures inter-modal and token-level dependencies to generate instrumental variables, which serve as auxiliary signals for disentanglement. 3) A learnable mask generator partitions the fused representation into initial causal and confounding

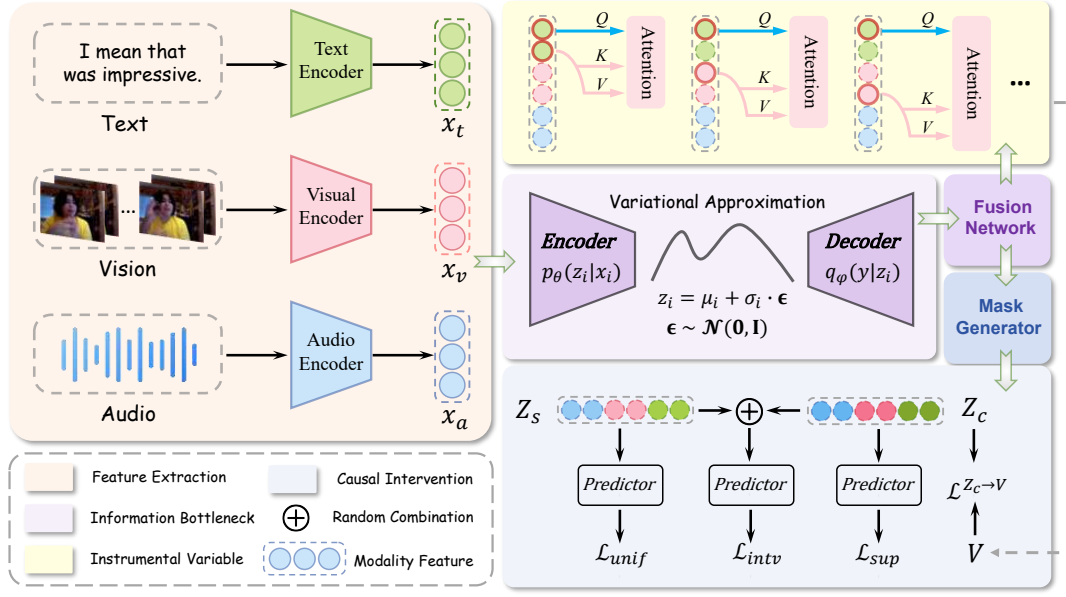


Figure 3: Overview of the CaMIB framework. During training, the components interact synergistically: stable representations are progressively reinforced in the causal branch, while unstable components are guided toward the confounding branch, leading to gradual disentanglement.

subrepresentations. 4) Finally, a backdoor adjustment strategy randomly recombines causal and confounding features to reduce their correlation. During training, these components interact synergistically: stable representations are progressively reinforced in the causal branch, while unstable components are guided toward the confounding branch, gradually disentangling the representations and ultimately yielding stable causal features for downstream tasks.

3.4 Information Bottleneck

To learn compact yet discriminative representations, we apply the IB principle to each unimodal input prior to fusion. The IB framework aims to compress each source state X_i into a latent bottleneck representation Z_i , where $i \in \{t, v, a\}$, by minimizing redundant information in the input while preserving the most relevant signals for the target label Y . Using mutual information (MI) to quantify the relevance between states, this trade-off is formulated as the following variational optimization problem:

$$\min_{p(z_i|x_i)} I(X_i; Z_i) - \beta I(Z_i; Y), \quad (1)$$

where $I(\cdot; \cdot)$ denotes the mutual information and the hyper-parameter β controls the balance between compression and predictive sufficiency. This objective enforces a trade-off between minimizing redundancy in the representation and maximizing its predictive power with respect to the target. By introducing parameterized networks, the term $I(X_i; Z_i)$ is modeled by a variational encoder $VAE_i : X_i \mapsto Z_i$ to extract information from the source, while the second term $I(Z_i; Y)$ is addressed by a decoder $q_\psi : Z_i \mapsto Y$ referring to the target state.

For the purpose of optimization, we utilize variational approximation to replace the intractable MI terms, yielding the following

tractable upper bound for each modality:

$$\mathcal{L}_{IB}^i = I(X_i; Z_i) - \beta I(Z_i; Y) \quad (2)$$

$$\leq \mathbb{E}_{p(x_i)} [\text{KL}(p_\theta(z_i|x_i) \| q(z_i))] - \beta \mathbb{E}_{p(x_i, y)} \mathbb{E}_{p(z_i|x_i)} [\log q_\psi(y|z_i)], \quad (3)$$

where the approximate posterior $p_\theta(z_i|x_i) \sim \mathcal{N}(\mu_i, \sigma_i^2)$ and the prior $q(z_i) \sim \mathcal{N}(0, I)$ follow Gaussian distributions. $\text{KL}(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence. Specifically, VAE_i maps the input X_i to the distribution parameters:

$$\mu_i, \sigma_i = VAE_i(x_i; \theta_{VAE_i}). \quad (4)$$

To enable gradient propagation and improve training efficiency, we leverage the reparameterization trick to sample the latent vector z_i :

$$z_i = \mu_i + \sigma_i \odot \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I). \quad (5)$$

Serving as a bridge that carries compactly relevant information between X_i and Y , the bottleneck latents Z_i effectively construct an informative space that filters out task-irrelevant noise while regulating the information flow across modalities. More detailed derivations are provided in the Appendix.

3.5 Modeling Causal Instrumental Variables

Statistically, causal structures are often governed by global semantic attributes rather than isolated unimodal features or local noise [9]. Consequently, the precise identification of causal components necessitates a holistic modeling of both cross-modal interactions and intra-modal contextual dependencies.

To this end, we introduce instrumental variables V [3, 54], designed to facilitate the extraction of causal features Z_c while effectively isolating the influence of confounding factors Z_s . Specifically,

let the compressed representations obtained from the IB module be concatenated to form a multimodal latent tensor $Z \in \mathbb{R}^{(m \cdot l) \times d}$, where m denotes the number of modalities, l the sequence length, and d the feature dimensionality. To capture the intricate inter-modal and token-level dependencies essential for the instrumental variable, we employ a self-attention mechanism. For each token $i \in \{1, \dots, m \cdot l\}$, the contextualized representation $\hat{v}_i$ is computed as:

$$\hat{v}_i = \sum_{j=1}^{m \cdot l} \frac{\exp(e_{ij})}{\sum_{n=1}^{m \cdot l} \exp(e_{in})} (z_j W_V), \quad e_{ij} = \frac{(z_i W_Q)(z_j W_K)^\top}{\sqrt{d}}, \quad (6)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d}$ are the learnable projection matrices for queries, keys, and values, respectively.

To derive a modality-agnostic yet context-aware instrumental signal, the resulting sequence $\hat{V} = \{\hat{v}_1, \dots, \hat{v}_{m \cdot l}\}$ is reshaped back into the modality-aware form $\mathbb{R}^{m \times l \times d}$ and subsequently aggregated across the modality dimension:

$$V = \sum_{k=1}^m \hat{V}_k, \quad V \in \mathbb{R}^{l \times d}, \quad (7)$$

where $\hat{V}_k$ denotes the contextualized feature matrix for the k -th modality. As a high-fidelity proxy encoding global dependencies across the multimodal space, V serves as a critical constraint for the subsequent extraction of causal features Z_c .

3.6 Causal Intervention

Given the compressed unimodal representations produced by the IB module, we first concatenate them to construct a joint multimodal latent tensor Z . To disentangle task-relevant semantics from spurious noise, Z is routed through two complementary pathways: a causal branch and a confounding branch. Rather than enforcing a rigid partition, this separation is realized as a gradual and adaptive disentanglement process during training.

Specifically, we introduce a learnable mask generator, implemented as a multilayer perceptron (MLP) followed by a sigmoid activation σ , to produce a continuous mask $M_c \in (0, 1)$ that indicates the likelihood of each element belonging to the causal feature space:

$$M_c = \sigma(\text{MLP}(Z)), \quad Z = \text{Concat}(Z_t, Z_o, Z_a). \quad (8)$$

The complementary confounding mask is defined as $M_s = 1 - M_c$. Using these masks, the multimodal representation Z is decomposed into a causal subrepresentation Z_c and a confounding subrepresentation Z_s via element-wise modulation:

$$Z_c = M_c \odot Z, \quad Z_s = M_s \odot Z. \quad (9)$$

At the early stages of training, this decomposition is inherently noisy and unstructured. To facilitate progressive disentanglement, we impose distinct supervision signals on the two branches and further apply causal intervention to stratify the confounding factors. For the causal branch, we encourage Z_c to align with a global instrumental variable V , thereby capturing modality-invariant dependencies, while simultaneously preserving sufficient information for accurate downstream prediction. The corresponding alignment

and task-specific objectives are defined as:

$$\mathcal{L}^{Z_c \rightarrow V} = \|Z_c - V\|^2, \quad \mathcal{L}_{sup} = -\frac{1}{N} \sum_{n=1}^N \log p(\hat{y}^n | z_c^n), \quad (10)$$

where $p(\hat{y}^n | z_c^n)$ denotes the predictive distribution based solely on Z_c . For classification tasks, $\mathcal{L}_{sup}$ corresponds to the cross-entropy loss, whereas for regression tasks, it reduces to the mean squared error. In contrast, the confounding branch is explicitly discouraged from encoding task-relevant information. Inspired by [61], we enforce its predictive distribution to collapse toward an uninformative uniform prior y_{unif} :

$$\mathcal{L}_{unif} = \frac{1}{N} \sum_{n=1}^N \text{KL}(p(\hat{y}^n | z_s^n) \| y_{unif}), \quad (11)$$

where $\text{KL}(\cdot \| \cdot)$ denotes the Kullback-Leibler divergence. In classification settings, y_{unif} is a uniform distribution over all classes; for regression, it is modeled as a zero-mean Gaussian with variance matched to the target distribution.

Finally, to effectively disentangle causal features from confounding features, we further introduce a causal intervention mechanism based on backdoor adjustment [37, 44]. The core idea is to enforce the invariance of causal predictions under diverse confounding contexts. Specifically, during training, we randomly recombine the causal representation z_c^n of a given instance with confounding features $z_s^{(k)}$ sampled from other instances within the mini-batch. The model is then required to produce consistent and accurate predictions based on the intervened representation z' , thereby preventing it from exploiting shortcut correlations and compelling it to rely exclusively on stable causal signals:

$$\mathcal{L}_{intv} = -\frac{1}{N \cdot |\hat{S}|} \sum_{n=1}^N \sum_{z_s^{(k)} \in \hat{S}} \log p(\hat{y}^n | z'), \quad z' = z_c^n + z_s^{(k)}, \quad (12)$$

where $\hat{S}$ denotes the set of confounding representations sampled from other instances within the current mini-batch, and $|\hat{S}|$ is its cardinality. This intervention strategy effectively simulates diverse confounding environments, encouraging the model to learn causal features that remain predictive regardless of the associated spurious variations.

3.7 Model Optimization

The overall training objective is formulated as a weighted combination of the above loss terms:

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_1 (\mathcal{L}^{Z_c \rightarrow V} + \mathcal{L}_{unif}) + \lambda_2 \mathcal{L}_{intv} + \sum_{i \in \{t, v, a\}} \mathcal{L}_{IB}^i, \quad (13)$$

where λ_1 and λ_2 are hyperparameters that balance the contributions of the disentanglement and intervention objectives. Under this unified optimization framework, the model exhibits a synergistic push-pull dynamic: invariant and generalizable semantics are progressively consolidated within the causal branch, while spurious correlations are systematically relegated to the confounding branch. As a result, robust and stable causal representations are learned.

Table 1: Comparison on the CMU-MOSI and CMU-MOSEI datasets. Acc2 and F1 scores are reported in two configurations: negative/non-negative (including zero) and negative/positive (excluding zero). [†] denotes that the results are taken from the original paper, while the remaining results are obtained from our own experiments. The best results are highlighted in bold, and the second-best results are underlined.

Method	CMU-MOSI					CMU-MOSEI				
	Acc2 (↑)	F1 (↑)	Acc7 (↑)	MAE (↓)	Corr (↑)	Acc2 (↑)	F1 (↑)	Acc7 (↑)	MAE (↓)	Corr (↑)
Self-MM [68] [AAAI'21]	83.12 / 84.75	83.09 / 84.78	46.15	0.719	0.788	79.90 / 84.45	80.57 / 84.50	53.23	0.531	0.768
ALMT [74] [EMNLP'23]	82.36 / 84.45	82.19 / 84.36	45.92	0.741	0.776	80.66 / 84.59	81.28 / 84.65	52.74	0.543	0.761
ConFEDE [67] [ACL'23]	83.24 / 85.12	83.20 / 85.15	43.29	0.728	0.784	81.21 / 85.65	81.78 / 85.55	52.72	0.538	0.772
DMD [22] [CVPR'23]	82.22 / 84.30	82.10 / 84.25	44.90	0.726	0.788	80.58 / 84.59	81.14 / 84.59	52.84	0.538	0.768
GLoMo [79] [ACM MM'24]	82.34 / 84.89	82.13 / 84.78	45.55	0.734	0.778	82.40 / 85.58	82.77 / 85.49	52.29	0.558	0.748
DLF [55] [AAAI'25]	82.22 / 84.76	82.05 / 84.69	46.65	0.731	0.787	81.95 / 85.20	82.40 / 85.17	52.14	0.543	0.764
DEVA [†] [58] [AAAI'25]	84.40 / 86.29	84.48 / 86.30	46.32	0.730	0.787	83.26 / 86.13	82.93 / 86.21	52.26	0.541	0.769
EMOE [10] [CVPR'25]	82.94 / 84.76	82.88 / 84.75	45.19	0.723	0.790	81.58 / 85.00	82.06 / 84.98	52.54	0.542	0.760
PSA-MF [†] [60] [AAAI'26]	83.67 / 86.43	83.30 / 86.19	46.50	0.686	0.807	83.76 / 86.30	84.21 / 86.28	55.00	<u>0.521</u>	0.774
C-MIB [31] [TMM'23]	85.40 / 87.18	85.34 / 87.16	<u>47.59</u>	0.650	0.840	83.72 / 86.57	84.10 / 86.56	<u>53.84</u>	0.526	0.779
ITHP [59] [ICLR'24]	86.13 / 88.24	86.00 / 88.17	46.28	0.654	0.844	82.30 / 86.24	82.88 / 86.32	51.58	0.556	0.781
KAN-MCP [29] [ACM MM'25]	<u>86.86</u> / <u>88.55</u>	<u>86.83</u> / <u>88.55</u>	47.30	<u>0.636</u>	<u>0.850</u>	<u>84.77</u> / <u>87.29</u>	<u>84.95</u> / <u>87.14</u>	53.52	0.522	0.793
CaMIB (Ours)	88.18 / 89.77	88.12 / 89.75	48.03	0.616	0.857	85.25 / 87.32	85.43 / 87.21	53.52	0.517	<u>0.788</u>

Table 2: Comparison on the UR-FUNNY dataset. [†] denotes that the results are taken from the original paper, while the remaining results are reported in [61].

Method	Context	Punchline	Accuracy (↑)
C-MFN [14] [EMNLP'19]		✓	64.47
C-MFN [14] [EMNLP'19]	✓	✓	65.23
Self-MM [68] [AAAI'21]		✓	68.77
FDMER [64] [ACM MM'22]		✓	70.55
DMD [22] [CVPR'23]		✓	68.70
AGM [†] [20] [ICCV'23]		✓	66.07
AtCAF [†] [18] [Inf. Fusion'25]	✓	✓	<u>72.13</u>
SuCI [61] [AAAI'25]		✓	70.92
DHMD [†] [23] [TPAMI'26]	✓	✓	71.30
CaMIB (Ours)	✓	✓	72.33

Table 3: Comparison on the MUSTARD dataset. [†] denotes that the results are taken from the original paper, while the remaining results are reported in [41].

Method	Context	Target	Accuracy (↑)
CIM-MTL [1] [NAACL'19]		✓	67.14
MAG-XLNet [42] [ACL'20]	✓	✓	74.72
BBFN [13] [ICMI'21]	✓	✓	71.42
MuLOT [41] [WACV'22]		✓	74.52
MuLOT [41] [WACV'22]	✓	✓	76.82
MIL [†] [77] [TAI'23]	✓	✓	76.30
ITHP [†] [59] [ICLR'24]	✓	✓	75.30
MOAC [†] [59] [WWW'25]	✓	✓	<u>80.88</u>
DHMD [†] [23] [TPAMI'26]	✓	✓	78.50
CaMIB (Ours)	✓	✓	82.35

3.8 Theoretical Analysis

In this subsection, we provide a theoretical analysis demonstrating that the proposed CaMIB framework effectively learns disentangled causal representations. By analyzing the optimization dynamics of the self-attention mechanism alongside the KL and MSE regularization constraints, we establish that the final learned representation Z satisfies three key properties: ❶ attention weights on causal features Z_c are strengthened, while those on confounding features Z_s are suppressed; ❷ the instrumental variable V is sensitive to Z_c but strictly independent of Z_s , mathematically guaranteeing $P(Z | V) \neq P(Z)$ and $P(Z_s | V) = P(Z_s)$; and ❸ the representation Z successfully blocks spurious shortcut paths, ensuring that predictions rely exclusively on causal signals, i.e., $P(Y | Z, V) = P(Y | Z_c)$. Ultimately, these properties theoretically guarantee the extraction of robust causal subrepresentations in multimodal learning. The detailed mathematical proofs and gradient derivations are deferred to Appendix.

4 Experiments

In this section, we evaluate CaMIB on four widely used multimodal datasets: the MSA datasets CMU-MOSI [71] and CMU-MOSEI [72],

the MHD dataset UR-FUNNY [14], and the MSD dataset MUSTARD [5]. To further assess model generalizability under distribution shifts, we also conduct experiments on the OOD variant of CMU-MOSI, with data splitting following [46]. For brevity, dataset information, evaluation metrics, baselines, implementation details, and additional results are provided in the Appendix.

4.1 Comparison with the state-of-the-art

Results on the MSA task. The results on the MSA task are reported in Table 1. Our proposed CaMIB achieves state-of-the-art performance on both the CMU-MOSI [71] and CMU-MOSEI [72] datasets. Specifically, CaMIB attains the highest Acc2 and F1 scores among all baselines, outperforming the second-best methods by 1.2%–1.3%. In addition, CaMIB significantly improves over existing methods in terms of MAE and Acc2 (including zero), highlighting its strong capability in predicting neutral sentiments. Furthermore, even when compared to models with the same backbone, such as C-MIB [31], ITHP [59], and KAN-MCP [29], CaMIB consistently demonstrates clear advantages. This superiority stems from its integration of the IB principle with causal theory, along with explicit

Table 4: Comparison on the OOD variant of the CMU-MOSI dataset. [†] indicates results reported in the original papers, [§] denotes results from our experiments, and the remaining results are taken from [46].

Method	Acc2 (↑)	F1 (↑)	Acc7 (↑)
MuT [51] [ACL'19]	75.00 / 76.72	74.75 / 76.52	29.80
MAG-BERT [42] [ACL'20]	75.57 / 77.28	75.52 / 77.26	39.85
MISA [15] [ACM MM'20]	75.90 / 77.39	75.8 / 77.35	38.05
Self-MM [68] [AAAI'21]	76.70 / 78.12	76.68 / 78.13	40.25
CLUE [46] [ACM MM'22]	78.75 / 79.94	78.75 / 79.93	41.75
GEAR [†] [45] [ACM MM'23]	80.47 / 82.10	80.44 / 82.09	-
MuDeF [†] [17] [TMM'24]	79.58 / 81.51	79.73 / 81.62	<u>42.92</u>
ITHP [§] [59] [ICLR'24]	79.50 / 81.25	79.49 / 81.26	41.25
KAN-MCP [§] [29] [ACM MM'25]	79.25 / 81.51	79.13 / 81.44	41.75
CaMIB (Ours)	82.75 / 84.38	82.73 / 84.38	44.50

Table 5: Ablation studies on the CMU-MOSI dataset.

	Acc7 (↑)	Acc2 (↑)	F1 (↑)	MAE (↓)
w/o Text-level IB	45.84	87.63	87.64	0.661
w/o Acoustic-level IB	46.57	88.55	88.54	0.628
w/o Visual-level IB	46.42	88.55	88.50	0.640
w/o Information Bottleneck	48.18	88.40	88.43	0.633
w/o Instrumental Variable	45.99	88.85	88.84	0.642
w/o KL Divergence Loss	48.47	88.85	88.85	0.633
w/ Mean Squared Error Loss	46.28	87.79	87.80	0.658
w/o Causal Intervention	46.86	87.94	87.96	0.626
w/o Random Combination	48.18	88.09	88.08	0.631
CaMIB (Ours)	48.03	89.77	89.75	0.616

modeling of global causality, enabling the learning of more stable and robust multimodal representations.

Results on the MHD and MSD tasks. To further evaluate the generalizability of CaMIB, we conduct experiments on the UR-FUNNY and MUSTARD datasets for MHD and MSD tasks, respectively. As shown in Table 2 and Table 3, CaMIB outperforms the latest state-of-the-art methods on the MHD task, including AtCAF [18] and DHMD [23]. For the MSD task, CaMIB achieves consistent improvements over all baseline models, surpassing the second-best method MOAC [32] by 1.47% in accuracy. Notably, compared to the IB-based method ITHP [59], CaMIB yields a significant improvement of 7.05%. Overall, CaMIB establishes new state-of-the-art results on both tasks, demonstrating its strong generalization capability across diverse MAC scenarios.

4.2 Out-of-Distribution Experiments

Table 4 presents the performance comparison between CaMIB and other methods under OOD test settings. Several observations can be drawn: ① Performance under OOD testing is consistently lower than that on the standard dataset for all methods, confirming that spurious correlations undermine generalization ability; ② Under OOD settings, CaMIB significantly outperforms baselines based on conventional multimodal techniques, with Acc2 and F1 improvements over ITHP [59] increasing from 2.1%/1.6% to 3.3%/3.2%, demonstrating the effectiveness of our confounding strategy; ③ Compared with causal-based methods, including CLUE [46], MuDeF [17], and GEAR [45], CaMIB achieves the best performance

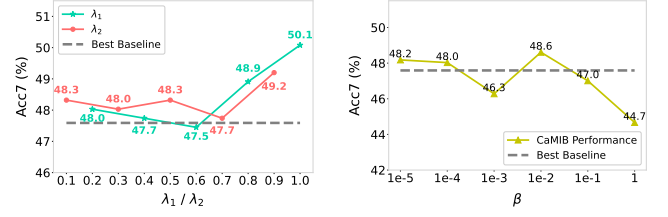


Figure 4: Parameter sensitivity analysis with respect to the disentanglement loss weight λ_1 , causal intervention loss weight λ_2 , and IB trade-off coefficient β . The gray dashed line indicates the best baseline performance.

across all evaluation metrics. While these methods mitigate spurious correlations through explicit causal intervention, counterfactual reasoning, or bias reweighting strategies, CaMIB bridges IB and causal theory while explicitly modeling global causality, enabling it to capture stable, task-relevant semantics shared across modalities. This leads to more effective suppression of dataset-specific biases and stronger OOD generalization.

4.3 Ablation study

We evaluate the impact of key components in CaMIB, and the results are reported in Table 5. Several observations can be drawn. **First**, IB improves performance, demonstrating the importance of applying IB to each input modality to learn compact, task-relevant representations while filtering out noise. Notably, applying IB to the textual modality yields the largest influence, as it is widely recognized as the dominant modality in the MSA task [33, 55]. **Second**, constructing instrumental variables is important for learning causal representations, particularly in improving fine-grained sentiment analysis. **Third**, removing the KL divergence loss leads to a performance drop, although the decrease is relatively moderate. This is because the confounding representation initially contains limited mutual information with the labels, which restricts the extent of disentanglement. Further evidence comes from an additional experiment: replacing the KL divergence loss with a mean squared error loss reduces CaMIB to a model with only instrumental variable constraints and insufficient disentanglement. As shown in Table 5, this results in a significant degradation across all metrics, highlighting the necessity of suppressing task-relevant information in Z_s . **Fourth**, removing the causal intervention module causes a substantial performance decline, indicating its critical role. Moreover, through random combination, each causal feature is paired with different confounding patterns. To further validate its effectiveness, we replace the “random combination” strategy with a deterministic “combination” variant that performs only ordered operations. The results show a performance drop, yet it still outperforms the model without causal intervention. This suggests that even partial disentanglement helps the model focus more on causal features.

4.4 Parameter Sensitivity Analysis

We analyze the sensitivity of our method to hyperparameters. Specifically, λ_1 controls the strength of disentanglement, λ_2 controls

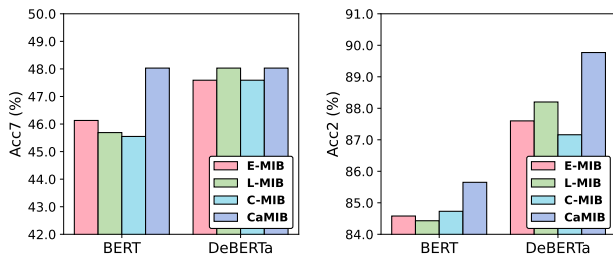


Figure 5: Comparison on the CMU-MOSI dataset. Results are reported using BERT and DeBERTa as text encoders.

Table 6: Comparison of model complexity on the CMU-MOSI dataset. Memory is measured with batch size 32, while training time per epoch is reported with batch size 8.

Method	FLOPs	Memory	Parameters	Training Time
C-MIB [31]	12.34 G	7.30 GB	190.93 M	16.09 s
ITHP [59]	12.04 G	7.13 GB	184.88 M	21.16 s
KAN-MCP [29]	12.99 G	12.98 GB	184.76 M	47.12 s
CaMIB (Ours)	11.82 G	7.75 GB	189.25 M	18.03 s

the strength of causal intervention, while β is the IB trade-off coefficient. The results are shown in Figure 4. The model is more stable with respect to λ_2 than λ_1 , as λ_2 aims to enhance the predictive consistency of multimodal representations under interventions. In contrast, the disentanglement process is relatively unstable: performance varies with λ_1 , although it remains within a controllable range (i.e., outperforming the best baseline in most cases). Overall, larger values of λ_1 and λ_2 tend to yield better performance. On the other hand, a large β significantly degrades performance. For example, when $\beta = 1$, the Acc7 score drops to 44.7%. Smaller values generally perform better, as the minimum information constraint does not dominate the training process while still preserving task-relevant information. Notably, CaMIB consistently outperforms the baselines across a wide range of settings, demonstrating strong robustness. Moreover, with careful tuning (e.g., $\lambda_1 = 1.0$), the model surpasses the default setting ($\lambda_1 = 0.2$), indicating further potential through fine-grained hyperparameter tuning.

4.5 More Analysis and Discussion

Comparison with MIB-based methods. We compare CaMIB with MIB [31] variants at different fusion stages, including early fusion (E-MIB), late fusion (L-MIB), and the complete framework (C-MIB). The results are shown in Figure 5, where CaMIB consistently outperforms these baselines with both BERT and DeBERTa as text encoders. Unlike conventional IB methods, which tend to overemphasize mutual information maximization while overlooking spurious correlations, CaMIB incorporates disentangled causal learning to effectively mitigate bias and improve generalization.

Discussion on the pre-trained language model. For the MSA task, following state-of-the-art methods [29, 59], we adopt DeBERTa [16] as the pre-trained text encoder. To demonstrate that the

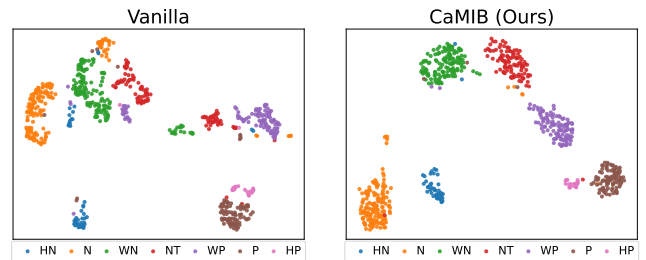


Figure 6: Visualization of the multimodal representations. HN: Highly Negative; N: Negative; WN: Weakly Negative; NT: Neutral; WP: Weak Positive; P: Positive; HP: Highly Positive.

effectiveness of our method is not limited to a specific encoder, we further evaluate it with BERT [8]. The results show that our model achieves strong performance under both encoders. Notably, with BERT as the text encoder, the Acc7 score reaches 48%, demonstrating a clear advantage. Additional results on the impact of different encoders across various baselines are provided in the Appendix.

Model complexity analysis. We present a comparison of computational complexity across different models, including FLOPs, memory usage, parameter count, and training time. As shown in Table 6, compared with C-MIB [31], ITHP [59], and KAN-MCP [29] under the same backbone, CaMIB achieves superior performance while maintaining competitive computational cost. Specifically, CaMIB requires fewer FLOPs than all baselines and exhibits comparable memory usage and parameter size. Although the training time is slightly higher than that of C-MIB, it remains significantly lower than KAN-MCP and comparable to ITHP. Overall, these results demonstrate that CaMIB strikes a favorable balance between efficiency and effectiveness, with limited computational overhead.

Visualization of the multimodal representations. To better understand the effectiveness of our method, we visualize the distribution of fused multimodal representations. As shown in Figure 6, compared with a vanilla multimodal model (using feature concatenation and MLP networks for fusion), the proposed CaMIB exhibits clearer separation among different emotion categories. This improvement can be attributed to the fact that CaMIB bridges the IB principle with causal learning and explicitly models global causality, leading to reduced noise and better disentanglement.

5 Conclusion

Most existing works predominantly follow the “learning to attend” paradigm, which can hinder out-of-distribution (OOD) generalization by conflating statistical shortcuts with true causal features. In this work, we propose a Causal Multimodal Information Bottleneck (CaMIB) framework that effectively captures global causal features while suppressing irrelevant noise. Extensive experiments on multiple MAC tasks and OOD test sets demonstrate that CaMIB achieves superior performance and strong robustness. Both theoretical and empirical analyses further validate the interpretability and solid causal foundations of our approach, offering a new perspective for multimodal learning.

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